Duration: 2 hours

**THE UNIVERSITY OF MANCHESTER**

# EART 22201 Igneous Minerals and Processes

**DATE:**

**TIME:**

Answer ALL questions in the booklets provided.

Clearly label which question each of your answers corresponds to. All writing must be LEGIBLE. Illegible documents cannot be marked.

There are 100 marks available on this paper. The number of marks for each question is shown in square brackets and may be used as a guide to how much time you should spend on each.

You are encouraged to use labelled sketches and diagrams to illustrate your answers.

**Consider time management as you prepare for this exam. Aim to budget about one minute per mark (100 minutes) – if you work to this timeframe you will have 20 minutes for checking and improving your work (or room to over-run your time budget on one or two questions).**

**Use the marks available per question and sub-question to guide how long to spend on each answer.**

- **A 2-mark question only needs a short answer – a couple sentences will be enough to hit the key points.**
- **An 8-, 12- or 15-mark question needs a more detailed, structured response over a few sentences or a couple of paragraphs. Consider whether a diagram (such as a graph or sketch) could be used to convey part of your answer.**

**If you have an individual support plan with extra time for invigilated assessments (e.g. 25% extra time), you should plan a time management strategy that will fit with your time allowance.**

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## Question 1 (20 marks)

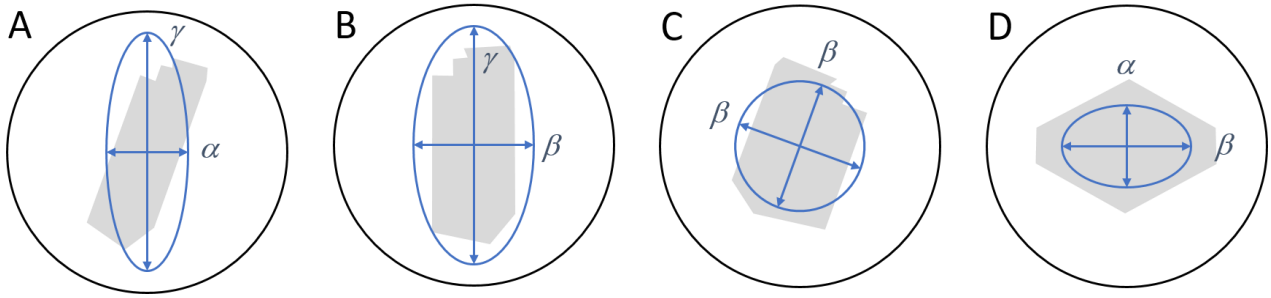
A silicate mineral belonging to the monoclinic crystal system has the following unit cell parameters:

**Unit cell:** a = 9.6 Å, b = 17.8 Å, c = 5.3 Å β = 104°
**Refractive indices:** α = 1.606, β = 1.620, γ= 1.627 β // y (β is parallel to y), γ^z = 15°
**Pleochroism:** α = colourless, β = lilac, γ= blue
**Cleavage:** {110}

a) What is the shape of the indicatrix for this mineral? [2]
Biaxial

b) What is the maximum value of birefringence in this mineral? [2]
0.021 (1.627 – 1.606)

The figure below shows four schematic images of the mineral viewed under a polarizing microscope, showing the orientation of the optical indicatrix. The grey colour indicates the position of the mineral on the microscope stage, NOT the colour observed under the microscope.




c) Which of these diagrams corresponds to:

i) A view of the mineral that, when the stage is rotated, will show blue-lilac pleochroism under plane-polarized light? [2]
B

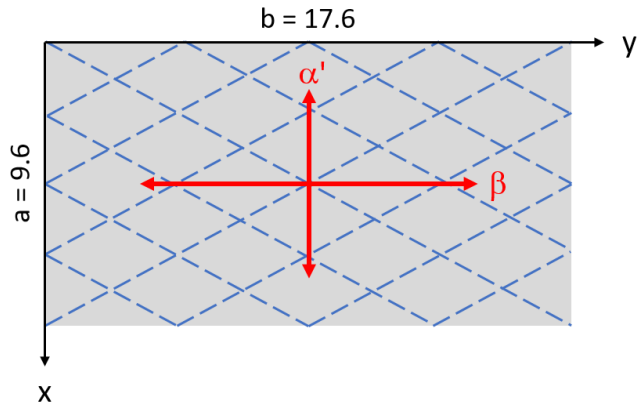
ii) A view of the mineral that will show inclined extinction with respect to the cleavage traces? [2]
A

iii) A view of the mineral that will show the interference colour corresponding to a birefringence of 0.014? [2]
D

iv) A view down one of the optic axes of the mineral? [2]

C

d) Draw a projection of the unit cell looking onto the (001) plane. Your diagram should include labelled crystallographic axes, and should show the shape of the unit cell, the orientation of the refractive indices, and the orientation of any cleavage traces. Use your diagram to determine whether this mineral is a pyroxene or an amphibole. [8]




Crystallographic axes: x and y [1]
Shape of the unit cell: a and b, correctly oriented (a along x, b along y) [2]
Refractive indices: β parallel to y, α (actually α') parallel to x [2]
Cleavage traces: two cleavages in {110} orientation one set of planes intersecting x and y at 1 unit cell repeat, and one set of planes intersecting x at 1 and y at -1 unit cell repeats. [2]

The angle between the cleavage traces is ~60°, so this is an amphibole. Award follow-through marks as long as mineral classification is consistent with the drawn orientation of cleavage traces. [1]

## Question 2 (20 marks)

a) Write one or two sentences to define each of the following terms:

i) Primary melt [2]

A basaltic melt with high MgO, generated by partial melting of mantle peridotite, that has not undergone any crystallization.

ii) Equilibrium crystallization [2]

Crystallization where the crystals and melt remain within the system (i.e., no melt or crystals are removed from the system at any point) and where the crystals and melt are always in chemical equilibrium.

iii) Mantle potential temperature [2]

The temperature that the mantle would have at the Earth's surface, if it ascended along an adiabat (i.e., following the convective geotherm) without melting.

b) What geochemical information would you use to tell the difference between mid-ocean ridge basalts and the basalts being erupted at Loihi seamount, a pre-shield stage submarine volcano in Hawaii? Your answer should refer to at least two different types of geochemical data. [8]

Mid-ocean ridge basalts (MORB) are tholeiitic. Basalts being erupted at Loihi and other pre-shield stage ocean islands (OIB) are alkali basalts.

1. Use TAS plot to distinguish between tholeiitic and alkali basalts. This requires major element compositions, specifically the $SiO_2$, $Na_2O$ and $K_2O$ contents. For a given $SiO_2$, content, the MORB tholeiitic basalts will have lower total alkali contents ($Na_2O+K_2O$) than Loihi alkali basalts.
2. Use trace elements. On a spider plot showing incompatible trace elements in order of incompatibility on the x axis, and normalized element concentration on the y axis, MORBs tend to have a slightly positive slope, whereas OIBs tend to have a slightly negative slope. The alkali basalts will be produced by a lower fraction of partial melting than the MORBs, so would expect the concentrations of incompatible trace elements to be higher in samples from Loihi than in typical MORB.
3. Isotopic compositions. MORB would be expected to have lower $^{87}Sr/^{86}Sr$ and higher $^{143}Nd/^{144}Nd$ than alkali basalts from Loihi. This is because MORB samples the depleted upper mantle, whereas ocean islands also sample a component of the primitive lower mantle.

Marks awarded for written explanation, or for illustrating the response on suitable graphs or diagrams. Diagrams may be schematic but must be sufficiently accurate to convey the necessary geochemical information.

<40: No answer, or incorrect information.

40-50: One type of geochemical data presented. Explanation may show misunderstandings or omissions.
50-60: Two types of geochemical data presented. Explanations may be superficial or may contain minor misunderstandings or omissions
60-70: Two types of geochemical data presented accurately. OR: One type of geochemical data presented with detail and insight, and one type of geochemical data described superficially.
70+: Two or more types of geochemical data presented, with critical insight into what these data might imply e.g. regarding partial melt fraction and/or mantle source.

c) Predict the igneous textures that might be observed if the magmas being generated at Loihi cooled and crystallized in a large magma chamber instead of erupting to the surface. Your answer should include a labelled sketch. [6]

An alkali basalt magma cooling and crystallizing in a large magma chamber would produce a gabbro. The essential minerals in an alkali gabbro would be plagioclase and clinopyroxene; type minerals are olivine and possibly some nepheline; might expect accessory oxides like magnetite.
The expected texture would be a coarse-grained cumulate. Early-crystallizing olivine is dense, so might sink to the floor of the chamber to form an olivine-rich layer. Alternatively, plagioclase might be the cumulus phase. Clinopyroxene could be a cumulus or intercumulus phase. If the cooling rate is sufficiently slow, the cpx might show exsolution lamellae.
The mineral assemblage, and composition of the minerals, will depend on height within the cooling intrusion: would expect higher modal proportions of mafic minerals at the base of the intrusion, and higher proportions of plagioclase towards the sandwich horizon.

An appropriate sketch should show a coarse-grained gabbro containing plagioclase + cpx (with or without other minerals). Any of cumulate/interstitial/poikilitic/ophitic texture would be appropriate; labels should be consistent with the sketch.

<40: No answer, or incorrect information.
40-50: Identifies an appropriate texture. Sketch is entirely schematic and/or does not correspond to the texture described. No information on expected mineralogy.
50-60: Identifies an appropriate texture and supported with a simple sketch. Expected mineralogy is mentioned, but with mistakes or omissions.
60-70: An appropriate texture is neatly sketched and labelled. Expected mineral assemblage is predicted and justified.
70+: An appropriate texture is neatly sketched and fully labelled. High first-class answer offers some insight into possible variations in mineralogy with height in the cooling intrusion.

Credit any appropriate remarks relating to the geochemical stratigraphy of layered mafic intrusions.

## Question 3 (30 marks)

a) A granite contains large megacrysts of alkali feldspar that are much larger than crystals of the other minerals in the rock. Suggest TWO possible ways in which this texture could form. For each case, explain what information (if any) the texture would provide about the order of crystallization. [10]

1. The megacrysts could be the intrusive equivalent of phenocrysts: they grew during an initial stage of very slow cooling in the magma. If this is the case, then megacrysts would tell us that alkali feldspar was the first mineral to begin crystallizing from the melt.
2. The megacrysts could reflect differences in crystal growth rates between alkali feldspar and the other minerals. Alkali feldspar is hard to nucleate but easy to grow, so it tends to form few but very large crystals. Other minerals are easier to nucleate but grow more slowly, so tend to form more but smaller crystals. If this is the case, then the crystal size would not tell us about the order of crystallization and other textural evidence would be needed to interpret the order of crystallization.

<40: No answer, or incorrect information.
40-50: One explanation offered. May show misunderstandings or omissions.
50-60: Two explanations offered. Answer is short and may contain minor misunderstandings or omissions.
60-70: Two explanations, both explained and linked to magmatic processes such as crystal growth. OR: One explanation given with detail and insight, and one explanation described only superficially.
70+: Both explanations fully developed, with critical insight into implications for interpretation of magmatic texture.

b) The alkali feldspars in Figure 3 show a characteristic intracrystalline texture. Give the name of this texture and briefly outline why it forms. Explain what information this texture provides about the cooling rate of the magma. [8]

Perthitic texture. It is an exsolution texture, caused when the alkali feldspar solid solution begins to separate into Na-rich and K-rich regions. This texture can only form under very slow cooling rates. If the cooling rate is fast, then no lamellae will form. The thickness of the exsolution lamellae is proportional to the cooling rate: the thicker the lamellae, the slower the cooling.

Credit any relevant remarks on the alkali feldspar solid solution, alkali feldspar phase diagram, kinetics of cation diffusion in minerals, etc.

<40: No answer, or incorrect information.
40-50: Correctly identifies an exsolution texture, but explanation contains significant misunderstandings or omissions.

50-60: Correctly identifies exsolution texture and its dependence on cooling rate. Explanation is short and may contain minor misunderstandings or omissions.
60-70: Exsolution and its dependence on cooling rate is fully explained, with reference to mineral chemistry.
70+: Critical insight into the process of exsolution; may comment on the breaking and re-forming of bonds and/or cation diffusion.

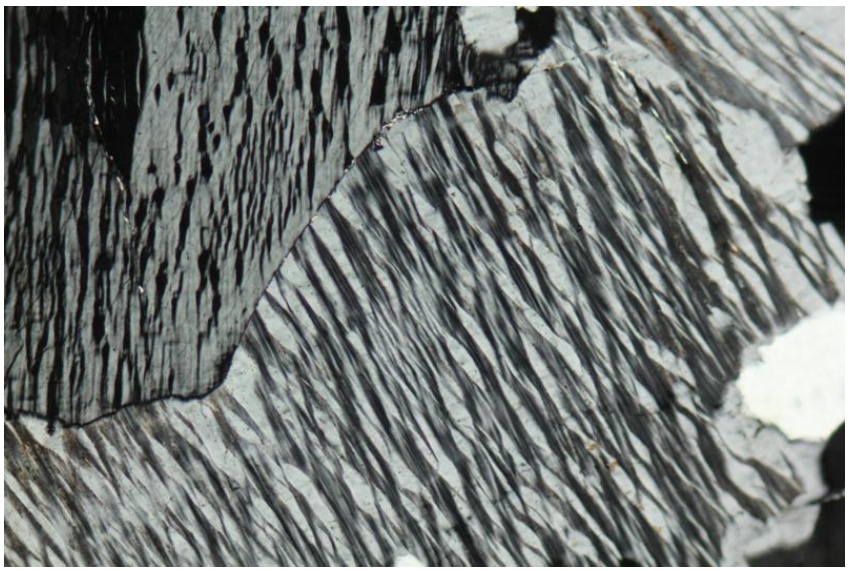


**Figure 3.** A characteristic intracrystalline texture in an orthoclase crystal from a granitic rock.

c) The major constituent minerals of a granite outcrop are quartz, alkali feldspar, plagioclase, and biotite. Based on this mineralogy, evaluate the likely origin of the magma, and suggest a tectonic setting in which this granite could have formed. Give reasons for your answer. [12]

This mineral assemblage is characteristic of I-type granite, meaning it is most likely derived from the remelting of igneous crustal rocks. Metallunimous I-type magmas are capable of crystallizing hornblende and biotite, which is consistent with the observed mineral assemblage. This sample can't be derived from melting meta-sedimentary crustal rocks, because this would produce a peraluminous (Al-rich) magma that would crystallize muscovite and other Al-rich minerals, which are not observed in this sample.

A likely tectonic setting is a continental margin such as the west coast of the USA. $H_2O$ release from the downgoing slab induces flux melting of peridotite in the mantle wedge, producing basalt which crystallizes at the base of the crust or intrudes into the crust. Partial melting of these originally intruded mafic magmas then produces the I-type magmas which can form large granite batholiths or erupt explosively at the arc volcanoes.

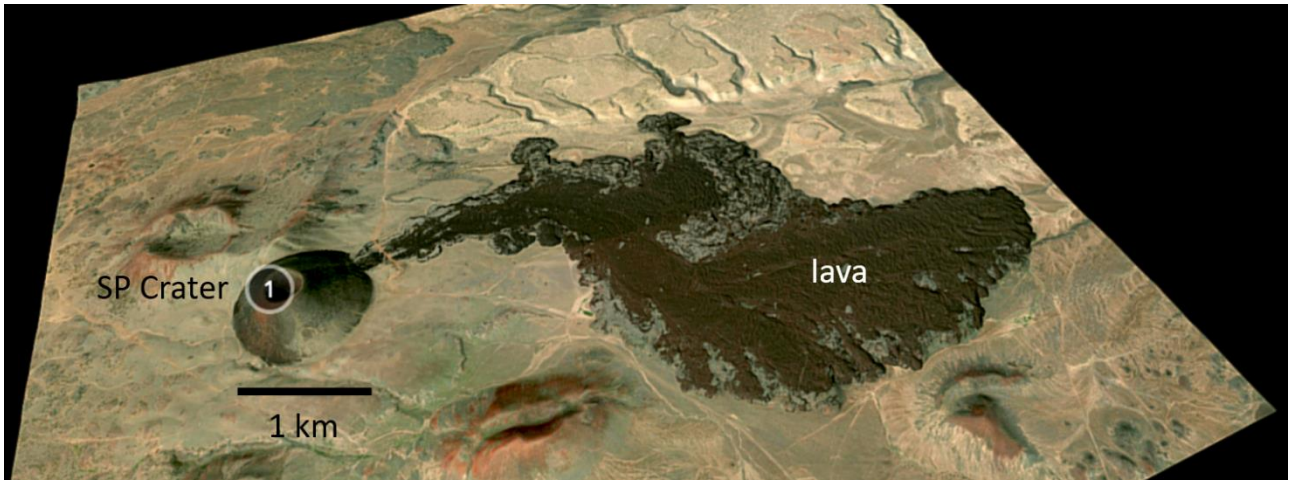
Credit any remarks about why any other tectonic setting is not likely. Award follow-through marks for comments on the tectonic setting if the granite has been classified incorrectly.

<40: No answer, or incorrect information.

40-50: Correctly classifies granite and/or correctly identifies the tectonic setting. Explanation contains significant omissions or misunderstandings.
50-60: Correctly classifies granite and correctly identifies the tectonic setting. Explanation is short, and may contain minor misunderstandings or omissions.
60-70: Formation of I-type granites explained in detail, covering all aspects of the question.
70+: Critical insight into granite formation; may remark on the melting processes occurring in the tectonic setting and/or offer comparison with S- or A-type granites.

## Question 4 (30 marks)

This question is based on SP Crater, a small volcano in the San Francisco Volcanic Field. The figure below shows a virtual outcrop of SP Crater and its associated lava flow.




The scoria cone at SP Crater is approximately 1 km in diameter and 200 m in height.

a) Calculate the approximate volume of the explosive volcanic products associated with SP Crater. The formula for the volume of a cone is given below. [5]

$$V = \frac{\pi r^2 h}{3}$$

Radius is 500 m, height is 200 m. Assume that SP Crater can be reduced to a simple cone geometry.
Calculated volume is 52,000,000 $m^3$ (0.052 $km^3$).
Given the assumptions and approximate dimensions, it is not appropriate to quote the answer to more significant figures.

b) Estimate the height of the volcanic plume that might have been associated with the eruption of SP Crater. Use the information in Table 1 to inform your answer, and briefly explain your working. [5]

The estimated volume of explosively erupted products makes this a VEI 3 eruption. VEI 3 eruptions should have column heights between 3 and 15 km, and erupt in Vulcanian style. Note that the volume of the lava isn't considered here, because the lava is an effusive eruption product (this is one of the limitations of the VEI as a quantitative scheme for classifying eruptions).

**Table 1.** The Volcanic Explosivity Index.

| VEI | Description | Plume height | Volume | Classification | Frequency | Example |
|---|---|---|---|---|---|---|
| 0 | non-explosive | <100 m | <10,000 $m^3$ | Hawaiian | constant/daily | Kilauea |
| 1 | gentle | 0.1-1 km | >10,000 $m^3$ | Hawaiian/Strombolian | daily | Stromboli |
| 2 | explosive | 1-5 km | >1,000,000 $m^3$ | Strombolian/Vulcanian | weekly | Whakaari 2019 |
| 3 | severe | 3-15 km | >0.01 $km^3$ | Vulcanian | few months | Anak Krakatau 2018 |
| 4 | cataclysmic | 10-25 km | >0.1 $km^3$ | Vulcanian/Plinian | ≥1 yr | Eyjafjallajökull 2010 |
| 5 | paroxysmal | >25 km | >1 $km^3$ | Plinian | ≥10 yr | Mount St Helens 1980 |
| 6 | colossal | >25 km | >10 $km^3$ | Plinian/ultra-Plinian | ≥100 yr | Krakatau 1883 |
| 7 | super-colossal | >25 km | >100 $km^3$ | ultra-Plinian | ≥1000 yr | Tambora 1815 |
| 8 | mega-colosal | >25 km | >1000 $km^3$ | ultra-Plinian | ≥10,000 yr | Yellowstone 2 Ma |

*Question 2 continues on the next page*

c) The San Francisco Volcanic Field is situated in a rift zone where lithospheric extension is occurring. What is the most likely mechanism of mantle melting that explains the magmatism in this location? Explain the process by which melting occurs, and state the tectonic setting that is most often associated with this mechanism of melting. [12]

The most likely melting mechanism is adiabatic decompression melting. This happens when the lithosphere is thinned, allowing the convecting asthenospheric geotherm to extend closer to the Earth's surface and potentially intersect the geotherm. This process most commonly occurs at mid-ocean ridges, where oceanic lithosphere is effectively thinned to zero due to plate separation. In a continental setting like the San Francisco Volcanic Field, the process of rifting serves to thin the lithosphere but doesn't reduce its thickness to zero (so we don't observe wholesale plate spreading). A first-class answer might comment that in this continental setting where rifting is only partial, the extent of partial melting may be quite small compared to a mid-ocean ridge.

<40: No answer, or incorrect information.
40-50: Able to identify adiabatic decompression melting; explanation contains significant misunderstandings or omissions.
50-60: Correctly identifies adiabatic decompression melting; explanation is short and may show minor misunderstandings or omissions.
60-70: Correctly identifies and explains the process of adiabatic decompression melting; answer covers all aspects of the question and is well explained.

70+: Critical insight into the melting process, offering points of comparison between melting at normal mid-ocean ridges and the San Francisco Volcanic Field.

d) The SP Crater lava flow contains olivine phenocrysts that show normal compositional zoning with high-forsterite cores and less forsteritic rims. Briefly explain the magma chamber process that might produce this zoning. [8]

Phenocrysts grow during slow cooling in the magma chamber. As crystallisation of olivine proceeds, the melt MgO content decreases and the composition of olivine in equilibrium with the melt becomes less forsteritic – this is governed by the olivine phase diagram. The normal zoning reflects the process of crystallization: the forsteritic olivine cores must have grown first, from a more primitive melt, and the more evolved olivine rim grew later. This is not fractional crystallization (otherwise all the phenocrysts would be removed as soon as they formed) but nor is it perfect equilibrium crystallization, because under true equilibrium conditions the whole crystal should be in chemical equilibrium with the melt.

A very good answer might comment that the forsteritic olivine cores could have grown in a different magma and been stirred into the SP Crater magma just before eruption. That is, the cores were carried into the magma chamber by a mafic recharge magma, and the rims grew later and are in equilibrium with the SP Crater melt composition.

<40: No answer, or incorrect information.
40-50: Identifies that olivine composition is governed by crystallization in the magma chamber; explanation contains significant misunderstandings or omissions.
50-60: Correctly identifies that olivine composition reflects progressive crystallization in the magma chamber; explanation is short and may show minor misunderstandings or omission.
60-70: Answer links olivine composition to the olivine phase diagram; most aspects of the question are well explained.
70+: Offers critical insights into the processes of equilibrium and fractional crystallization. May propose alternative explanations for the olivine zoning.

**END OF EXAMINATION PAPER**